

# Paper Data and Demo Outputs

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*Numerical records for device-free practice; not an interpretation key*

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

## How to use these outputs

These are outputs from a locally executed demonstration using course pseudonym demo-001. They allow the same questions to be answered without a device. Students with their own pseudonym may obtain different random values. Do not copy a numerical interpretation without understanding it; the scientific questions still require your own explanation. Every generated dataset is synthetic; limited real excerpts/cards are separately identified. This file is not an empirical research-results report.

For a paper D2, use the W09 split counts and metrics below. There are 12 synthetic donors, 6 rows each, 8 features. Within-donor training contains 4 rows/donor and testing 2 rows/donor. The group split holds out 4 donors entirely. Scaling is fitted on training data only; donor predictions use majority vote, with a 50:50 tie assigned to class 1. The 100 null permutations shuffle donor labels within the fixed training/test partitions, preserving their class counts. Explain the resulting evidence boundaries; a computer is not required to identify the design flaw.

### W01 | Demonstration output

data\_status: SYNTHETIC

sequence: ACCAATGGTGGCACTTAAAGCCCAAACCTCT

GC\_known: 0.4666666666666667

### W02 | Demonstration output

data\_status: SYNTHETIC

selected: GTACG

length: 5

reference\_build: toy\_contig; no human assembly

inclusive:

```
[
  3,
  7
]
```

### W03 | Demonstration output

data\_status: SYNTHETIC

GCNN\_bug: 0.5

GCNN\_fixed: 1.0

### W04 | Demonstration output

data\_status: SYNTHETIC

fixed\_score: 5

fixed\_identity: 0.75

best\_local\_score: 5

## W05 | Demonstration output

data\_status: SYNTHETIC scenario; hypothetical cost model

designs:

```
[
  {
    "name": "A",
    "n_per_condition": 2,
    "million_pairs_per_sample": 80,
    "samples": 4,
    "cost": 3800,
    "usable_million_pairs": 64.0,
    "within_budget": true
  },
  {
    "name": "B",
    "n_per_condition": 4,
    "million_pairs_per_sample": 30,
    "samples": 8,
    "cost": 3600,
    "usable_million_pairs": 24.0,
    "within_budget": true
  },
  {
    "name": "C",
    "n_per_condition": 6,
    "million_pairs_per_sample": 15,
    "samples": 12,
    "cost": 3600,
    "usable_million_pairs": 12.0,
    "within_budget": true
  }
]
```

nominal\_toy\_DNA\_depth: 60.0

usable\_toy\_DNA\_depth: 48.0

## W06 | Demonstration output

data\_status: SYNTHETIC all-null p-values; independent Uniform(0,1)

worked\_BH:

```
[
```

```
0.004,  
0.02,  
0.05333333333333334,  
0.2  
]  
  
mean_raw_false_positives: 5.026  
  
fraction_families_with_BH_discovery: 0.064  
  
mean_BH_discoveries: 0.068
```

## W07 | Demonstration output

data\_status: REAL 2-gene excerpt for metadata exercise; SEPARATE synthetic TPM example

matched\_conditions:

```
[  
  "untreated",  
  "untreated",  
  "untreated",  
  "untreated",  
  "treated",  
  "treated",  
  "treated"  
]
```

toy\_TPM:

```
[  
  500000.0,  
  500000.0  
]
```

## W08 | Demonstration output

data\_status: FICTIONAL claims for evidence audit

claims:

```
[  
  "p = 0.03 means a 3% probability that the null hypothesis is true.",  
  "Five hundred cells from one donor are five hundred independent donors.",  
  "A pLDDT of 95 establishes strong binding affinity.",  
  "A file header reading chr1 proves that the reference is GRCh38.",  
  "For the toy complete two-gene library, the computed TPM values sum to one million.",  
  "Removing names from a genome file guarantees anonymous public sharing."  
]
```

## W09 | Demonstration output

data\_status: SYNTHETIC random donor labels and donor signatures

within\_donor\_split:

```
{  
  "row_accuracy": 1.0,  
  "donor_accuracy": 1.0,  
  "test_donors": 12,  
  "test_rows": 24  
}
```

new\_donor\_split:

```
{  
  "row_accuracy": 0.75,  
  "donor_accuracy": 0.75,  
  "test_donors": 4,  
  "test_rows": 24  
}
```

conditional\_permutation\_mean: 0.495

permutation\_min\_max:

```
[  
  0.0,  
  1.0  
]
```

held\_out\_donors:

```
[  
  7,  
  8,  
  10,  
  11  
]
```

## W10 | Demonstration output

data\_status: SYNTHETIC integer counts

cells: 32

pseudobulk\_groups: 8

donors\_per\_condition: 2

G1\_total: 147

## W11 | Demonstration output

data\_status: REAL bibliographic evidence cards; proposed evaluation not executed

scorecard:

```
{  
  "task": "cell-type classification in unseen donors",  
  "baseline": "simple classifier on selected genes",  
}
```

```

"split": "hold out donors before preprocessing",
"adaptation": "declare zero-shot OR fine-tuned",
"metric": "per-class recall and balanced accuracy",
"missing": "external-cohort performance and uncertainty"
}

```

## W12 | Demonstration output

data\_status: REAL record summary plus SEPARATE synthetic confidence arrays

low\_confidence\_positions\_1 based:

```

[
  6,
  7,
  8
]

```

synthetic\_mean\_pLDDT: 80.75

synthetic\_PAE\_blocks:

```

[
  [
    2,
    18
  ],
  [
    20,
    3
  ]
]

```

## W13 | Demonstration output

data\_status: FICTIONAL nonclinical variant cards; not real genes or patients

cards:

```

[
  {
    "id": "TEACH-A",
    "gene": "FICTIONAL_GENE1",
    "condition": "fictional recessive disorder",
    "classification": "uncertain significance",
    "review": "single submitter with criteria",
    "evidence": "rare; computational tools disagree; segregation unknown"
  },
  {
    "id": "TEACH-B",
    "gene": "FICTIONAL_GENE2",
    "condition": "fictional dominant disorder",

```

```

    "classification": "conflicting submissions",
    "review": "criteria provided, conflicting classifications",
    "evidence": "older pathogenic assertion; later likely benign assertion; full records needed"
  },
  {
    "id": "TEACH-C",
    "gene": "FICTIONAL_GENE3",
    "condition": "fictional disorder",
    "classification": "pathogenic",
    "review": "expert panel",
    "evidence": "classification is stated for a particular disease and inheritance context"
  }
]

```

## W14 | Demonstration output

data\_status: FICTIONAL data-governance scenarios

scenarios:

```

[
  {
    "id": "A",
    "data": "fully synthetic course sequences generated by code",
    "destination": "class repository"
  },
  {
    "id": "B",
    "data": "patient genome with names removed",
    "destination": "public chatbot"
  },
  {
    "id": "C",
    "data": "controlled-access health data",
    "destination": "personal overseas cloud drive"
  },
  {
    "id": "D",
    "data": "published aggregate plot",
    "destination": "report with citation and licence check"
  },
  {
    "id": "E",
    "data": "classmates' identifiable assignments",
    "destination": "external AI grading service"
  }
]

```

## Inspection records

Toy FASTA: ACGTACGTAA. BED: toy\_contig, start2, end7. GTF exon: positions3through7, positive strand. Toy VCF: position4, REF T, ALT C. These describe a synthetic contig, not a human assembly.

Real two-gene count excerpt and sample metadata are in the Data folder. Only the first two genes displayed in the DESeq2 vignette are included. Source[S10/S25]. Treating them as a complete biological library is not permitted.